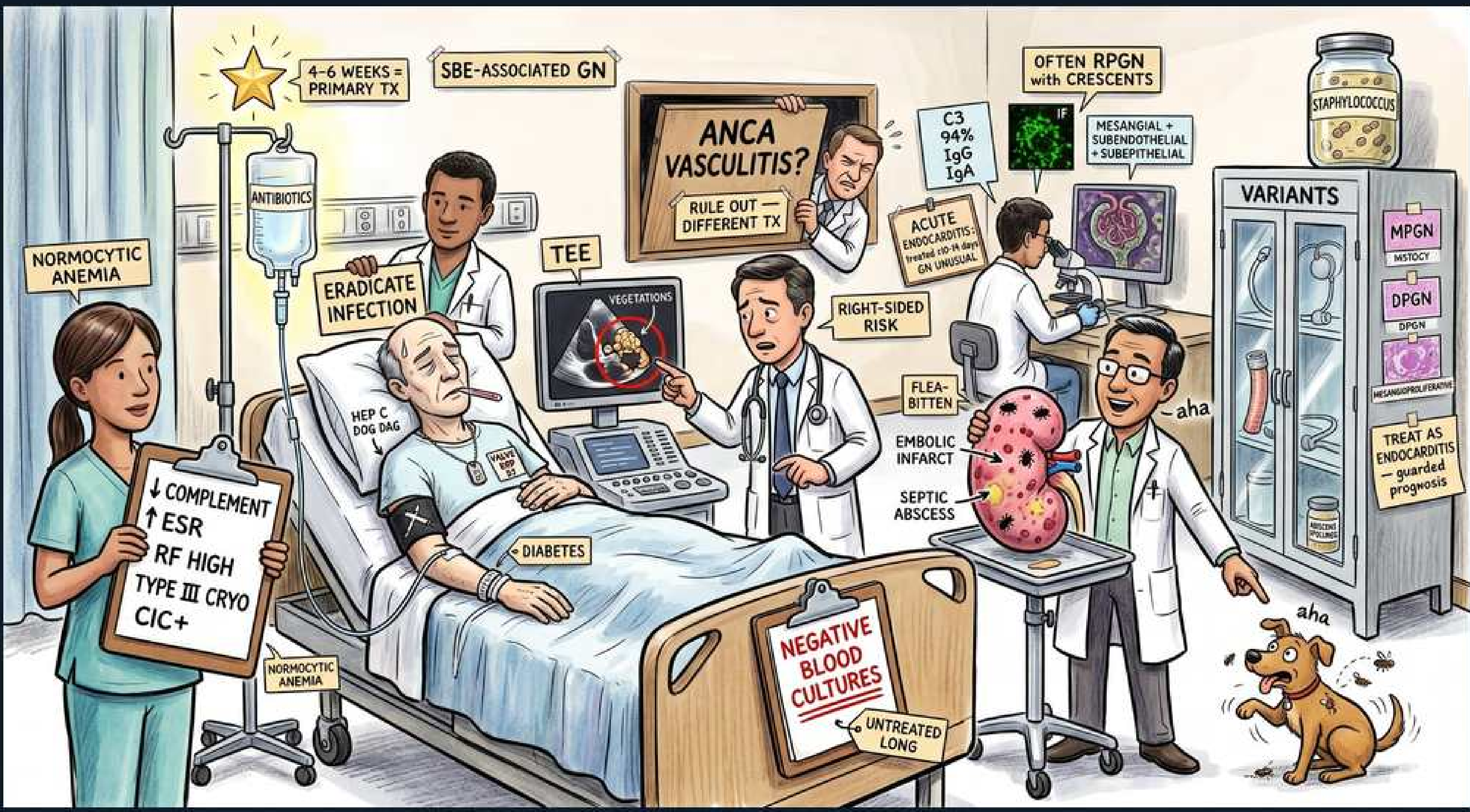
Glomerular (Nephritic) — Endocarditis-Associated GN



1. SBE-associated GN

WHAT YOU SEE

A banner over the hospital room reading SBE-ASSOCIATED GN — the title card for endocarditis-driven immune-complex nephritis.

WHAT IT ENCODES

Endocarditis-associated (classically subacute bacterial endocarditis, SBE) glomerulonephritis is an immune-complex nephritic GN driven by chronic intravascular infection. Persistent bacteremia generates circulating immune complexes that deposit in glomeruli, producing hematuria, RBC casts, proteinuria, and AKI. It is an infection-related GN and behaves like a 'low-complement nephritic' picture. Key teaching point: the kidney recovers when the infection is cured, not when you immunosuppress.

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2. Eradicate infection = primary Tx

WHAT YOU SEE

An ERADICATE INFECTION antibiotic IV drip tagged 4-6 WEEKS = PRIMARY TX wearing a gold star — the gold-star antibiotic bag marks curing the infection as the first-line treatment.

WHAT IT ENCODES

The PRIMARY treatment is eradication of the underlying infection with prolonged IV antibiotics (typically 4–6 weeks, organism-directed), with valve surgery if indicated — NOT immunosuppression. Renal function generally improves once bacteremia clears. Steroids/cyclophosphamide are reserved for severe crescentic disease only after the infection is controlled, because immunosuppressing an active infection can be catastrophic.

BOARD TRAP

Start cyclophosphamide or steroids first — you must treat the infection first; reflexive immunosuppression in active endocarditis can worsen sepsis

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3. Low complement

WHAT YOU SEE

A nurse holding a card reading ↓ COMPLEMENT, ↑ ESR, RF HIGH, TYPE III CRYO, CIC+ — the down-arrow on complement shows immune complexes are consuming it (a low-complement nephritic picture).

WHAT IT ENCODES

Serum complement is LOW (especially C3) because circulating immune complexes consume complement — placing endocarditis GN among the classic low-complement nephritic causes (with PSGN, lupus, MPGN, and cryoglobulinemia). ESR and CRP are elevated and rheumatoid factor may be positive. Tracking complement normalization can mirror successful infection clearance.

BOARD TRAP

Normal complement — complement is LOW in endocarditis-associated GN; normal complement instead suggests anti-GBM or ANCA/pauci-immune disease

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4. Type III cryoglobulins / CIC

WHAT YOU SEE

The same lab card listing TYPE III CRYO and CIC+ — the polyclonal cryoglobulins and circulating immune complexes are the mechanistic bridge from the infected valve to the glomerulus.

WHAT IT ENCODES

Chronic antigenemia drives formation of circulating immune complexes (CIC) and type III (polyclonal) cryoglobulins that lodge in glomerular capillaries and activate complement. This is the mechanistic link between the heart infection and the kidney lesion. Cryoglobulinemic features (purpura, arthralgias) can accompany the renal picture and reinforce the immune-complex mechanism.

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5. ANCA may be positive

WHAT YOU SEE

A figure clutching his head beside a sign reading ANCA VASCULITIS? — RULE OUT, DIFFERENT TX — the confused doctor depicts endocarditis faking a positive ANCA, a trap with opposite treatment.

WHAT IT ENCODES

Endocarditis can produce a false-positive ANCA (often PR3/c-ANCA) and clinically mimic ANCA-associated vasculitis — but the treatments are opposite (antibiotics vs. immunosuppression). ALWAYS draw blood cultures and get an echocardiogram before immunosuppressing an apparent ANCA vasculitis, because missing endocarditis and giving cyclophosphamide is a classic, dangerous board error. Low complement and positive cultures point toward endocarditis rather than primary vasculitis.

BOARD TRAP

ANCA-positive automatically means give immunosuppression — rule out endocarditis first; endocarditis-associated ANCA is treated with antibiotics, and immunosuppression would be harmful

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6. Mesangial/subendothelial/subepithelial deposits

WHAT YOU SEE

A microscopy/IF panel reading C3 94%, IgG IgM with deposits in MESANGIAL + SUBENDOTHELIAL + SUBEPITHELIAL sites — the granular C3-dominant staining IS the immune-complex deposition.

WHAT IT ENCODES

Immunofluorescence shows C3 (in the vast majority) often with IgG and IgM in a granular pattern, and electron microscopy reveals deposits in mesangial, subendothelial, and sometimes subepithelial locations. The picture can resemble a proliferative or membranoproliferative (MPGN) pattern. Granular ('lumpy-bumpy') staining confirms the immune-complex nature and separates it from the linear IgG of anti-GBM.

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7. Often RPGN with crescents

WHAT YOU SEE

A sign reading OFTEN RPGN WITH CRESCENTS — flags that severe cases form crescents and progress rapidly.

WHAT IT ENCODES

Severe cases present as crescentic, rapidly progressive GN with a steep rise in creatinine. Even then, the first move is aggressive infection control; immunosuppression is added only for biopsy-proven crescentic disease that fails to improve with antibiotics. The presence of crescents worsens renal prognosis and warrants biopsy.

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8. TEE — find vegetations

WHAT YOU SEE

An ultrasound screen showing a heart with circled VEGETATIONS, labelled TEE — the echo image of valve clumps IS transesophageal echo finding the vegetations.

WHAT IT ENCODES

Echocardiography identifies valvular vegetations; transesophageal echo (TEE) is far more sensitive than transthoracic (TTE) — roughly 90%+ vs ~60–70% — and is the test of choice when suspicion is high or TTE is negative. Vegetations plus positive cultures satisfy major Duke criteria. In a nephritic patient with fever and a new murmur, get a TEE.

BOARD TRAP

A negative transthoracic echo rules out endocarditis — it does not; proceed to TEE, which is much more sensitive

9. Blood cultures

WHAT YOU SEE

A red sign reading NEGATIVE BLOOD CULTURES / UNTREATED LONG — warns that pretreatment or fastidious organisms can give falsely negative cultures.

WHAT IT ENCODES

Multiple sets of blood cultures (ideally 3 sets from separate sites before antibiotics) identify the organism and are a major Duke criterion. Cultures may be falsely negative if the patient was pretreated with antibiotics or infected by fastidious/HACEK or Bartonella/Coxiella organisms — consider serologies in 'culture-negative endocarditis.' Always culture before starting empiric therapy when feasible.

10. Staphylococcus organism

WHAT YOU SEE

A specimen jar labelled STAPHYLOCOCCUS — the bug in the jar is the leading cause of acute endocarditis.

WHAT IT ENCODES

Staphylococcus aureus is the leading cause of acute endocarditis and is especially common in IV drug users and prosthetic valves; viridans streptococci classically cause the subacute (SBE) form on previously abnormal valves. Organism dictates antibiotic choice and surgical urgency. S. aureus endocarditis tends to be aggressive with rapid valve destruction and embolic complications.

11. Right-sided / IVDU risk

WHAT YOU SEE

A speckled FLEA-BITTEN kidney with EMBOLIC INFARCT and SEPTIC ABSCESS, tagged RIGHT-SIDED RISK — the dot-peppered kidney depicts embolic infarcts, with right-sided IVDU disease showering the lungs.

WHAT IT ENCODES

IV drug use predisposes to RIGHT-sided (tricuspid) endocarditis, which showers septic emboli to the LUNGS rather than the systemic circulation. Left-sided disease embolizes systemically, producing renal infarcts, the classic 'flea-bitten' kidney, and septic abscesses. So renal injury in endocarditis comes from two mechanisms: immune-complex GN and embolic infarction.

BOARD TRAP

IVDU endocarditis embolizes to the kidneys/brain — right-sided (tricuspid) disease from IVDU embolizes to the LUNGS, not the systemic circulation

12. Normocytic anemia

WHAT YOU SEE

A nurse card reading NORMOCYTIC ANEMIA — the chronic-infection anemia of chronic disease.

WHAT IT ENCODES

Chronic infection produces a normocytic, normochromic anemia of chronic disease along with elevated ESR/CRP. Other supportive subacute findings include splenomegaly, Roth spots, Osler nodes, Janeway lesions, and splinter hemorrhages. These chronic-infection markers help distinguish subacute endocarditis from an acute primary GN.

13. Treat as endocarditis = better prognosis

WHAT YOU SEE

A cabinet of VARIANTS (MPGN, DPGN) with a sign reading TREAT AS ENDOCARDITIS — the message that curing the infection (not immunosuppressing) gives the better outcome.

WHAT IT ENCODES

Renal recovery parallels infection control, so framing and managing the case as endocarditis (cure the infection ± valve surgery) yields better outcomes than mislabeling it as a primary vasculitis and immunosuppressing. Histologic variants include diffuse proliferative (DPGN) and membranoproliferative (MPGN) patterns. The unifying exam message: find and treat the infection.